18.5 Bicycle Lane Closed – Bicycle Detour

Purpose:

This layout shows a bicycle lane closure where a bicycle detour is required.

A bicycle detour may be necessary where it is not possible to maintain sufficient roadway surface or width for cyclists through the work zone.

In this layout, cyclists are instructed to detour to a different route in advance of the construction area. Where possible, the detour should begin at a location where the construction is within sight so that cyclists can see the reason for the detour.

Standard:

- A series of Bicycle Detour B-C-004 signs with appropriate directions shall be used along the detour.

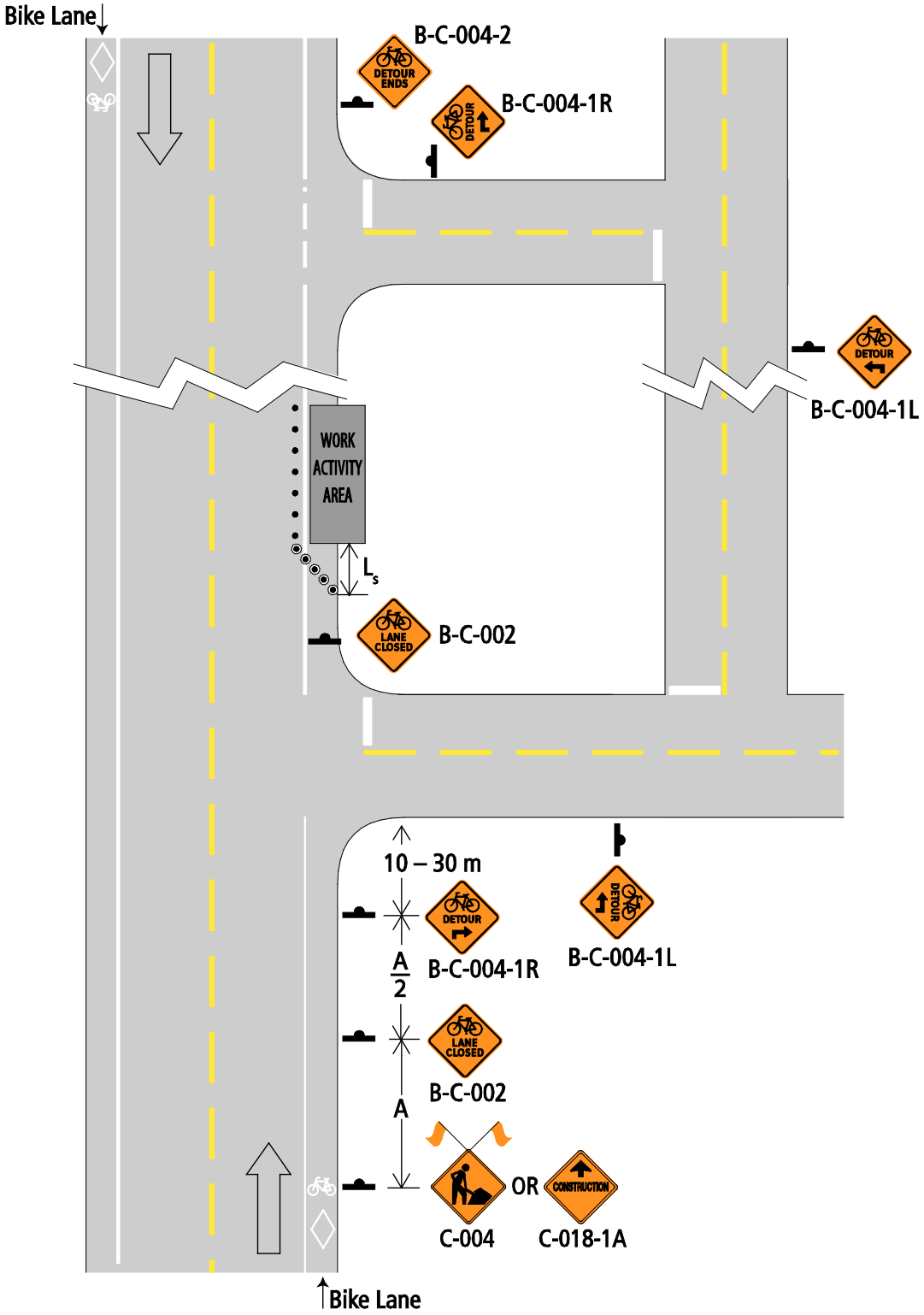
Guidance:

- Bicycle Detour Ahead B-C-004-1A signs should be used in advance of intersections along the route where the cyclist needs to continue straight ahead to stay on the detour route.
- Detours in which cyclists have to make left turns or cross arterial roadways without a signal or push-button crosswalk should be avoided.
- Detours for cyclists may not be possible on rural highways because there are few alternative routes, and detours may be lengthy. Other provisions should be considered, such as providing a shuttle service for cyclists.

Options:

- Bicycle Detour Ahead C-004-1A signs may be used along the detour route for confirmation.
- Where space allows, an additional Crew Working Ahead C-004 sign may be used in advance of the closure at a distance of one-half of Table B Distance A.

Figure 18.5: Bicycle Lane Closed – Bicycle Detour



18.6 Bicycle Lane Closed – Dismount and Walk

Purpose:

It is preferable for cyclists to continue cycling when work occurs on a bicycle route but certain conditions may prevent this, such as the nature of the road alignment or surface.

This layout shows a bicycle lane closure where cyclists must dismount and walk around the closure. This may be preferable to a bicycle detour if the closure length is relatively short (less than 50 metres). Cyclists will usually be dismounting and walking along a sidewalk. If a sidewalk is not present, the walking route must be marked with channelizing devices.

The layout illustrates a bicycle lane closure in a low-volume residential neighbourhood as a common example of where the "dismount and walk" requirement would probably occur. It is primarily intended to illustrate bicycle-related signage, but additional signage that is not bicycle-specific may also be required.

Standard:

- In advance of the work activity area, a confirmatory Bike Lane Closed B-C-002 sign with Cyclists Stop and Dismount B-R-101-Tb tab shall be used to direct cyclists to dismount.
- A Walk Bicycle B-R-101-2 sign shall be placed at the beginning of the area where cyclists shall walk. If a sidewalk is present, an On Sidewalk B-R-101-Tc tab should be used with the B-R-101-2 sign.

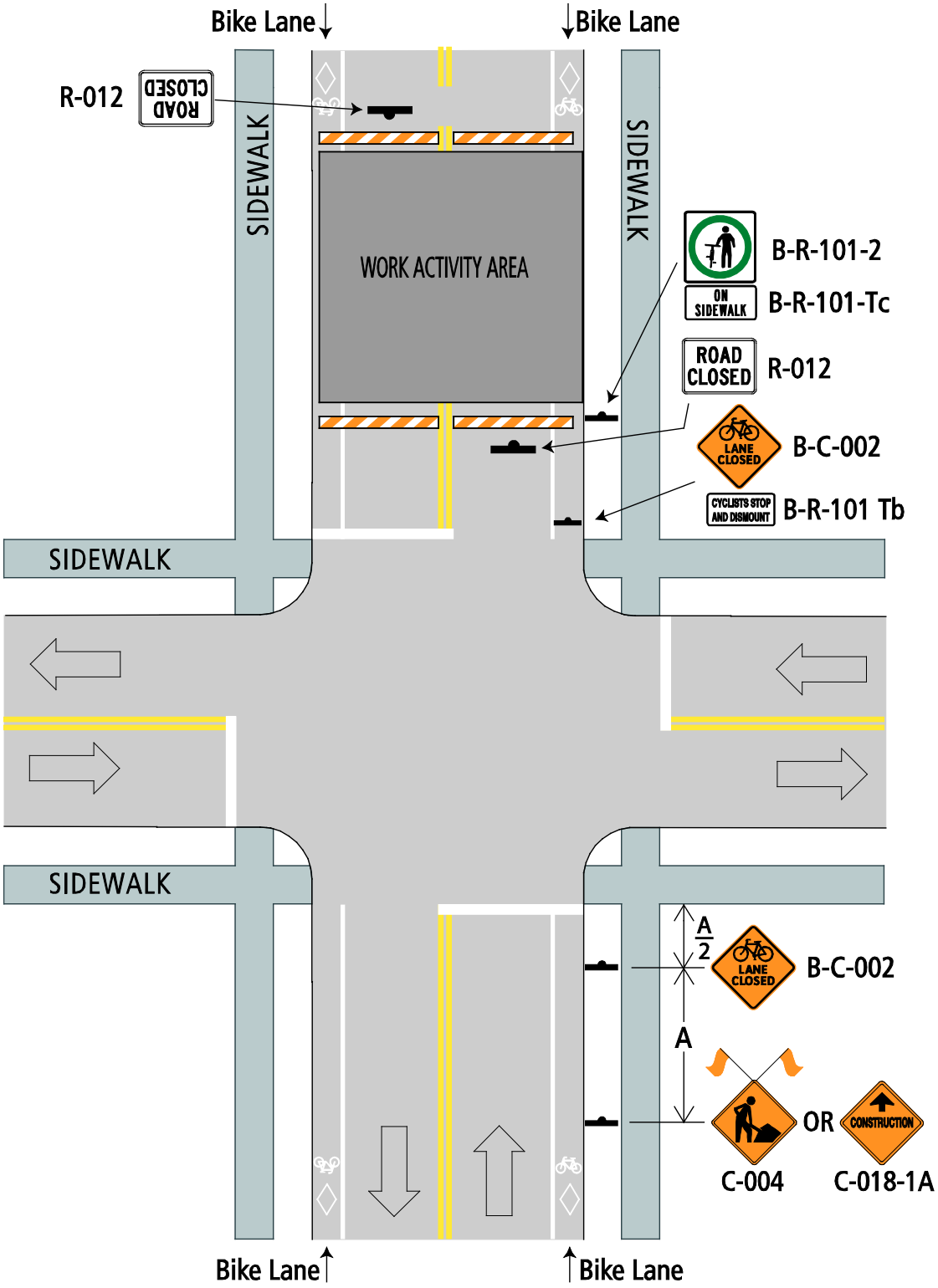
Guidance:

- If the reason for dismounting is not obvious, additional signage identifying the hazard should be installed (e.g., Loose Gravel C-015 sign).

Options:

- If cyclists are ignoring the requirement to dismount, a worker or Traffic Control Person near the dismount point may provide guidance to cyclists.

Figure 18.6: Bicycle Lane Closed – Dismount and Walk



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PART B– TRAFFIC CONTROL

Section 19: Traffic Control Layouts – Other Scenarios

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19.8 Emergency Response 19-28

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PART B – TRAFFIC CONTROL

Section 19: Traffic Control Layouts – Other Scenarios

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

19.1 Tow Truck Recovery Operations

Purpose:

This section outlines the key issues that should be considered to maintain public and worker safety when recovering disabled vehicles on or near the roadway.

The effects of towing operations on traffic may be similar to the effects of construction activities so the appropriate traffic control layouts in Sections 7 to 18 should be used for tow truck recovery operations.

Standard:

All work vehicles shall have 360-degree flashing lights and 4-way flashers.

- When a vehicle incident is affecting the flow of traffic, or if recovery operations could affect traffic flow for more than 15 minutes, additional traffic control is required.
- The most appropriate traffic control layouts shall be applied for vehicle recovery operations which exceed 15 minutes in duration. These may include, but are not limited to, the following scenarios:
  - A disabled vehicle and tow truck are on the shoulder, and recovery operations will not impact traffic (see *Section 7.5: Work on Shoulder*).
  - A disabled vehicle and/or tow truck are on the shoulder but recovery operations are encroaching, or will encroach, into the travel lane (see *Section 7.7: Roadside Work – Encroachment into Travel Lane*).
  - A disabled vehicle and/or tow truck are in one lane of a two-lane, two-way roadway. Operators will have to determine which layout is most appropriate based on the anticipated duration of the recovery operations (see *Section 7.3: Emergent Work* or *Section 7.8: Lane Closure with TCPs – Single Lane Alternating*).
  - In *Section 8: Traffic Control Layouts – Multilane Undivided Roadways,* see the Right Lane Closed, Left Lane Closed, and Centre Lane Closure subsections.
  - In *Section 9: Traffic Control Layouts – Multilane Divided Roadways,* see the corresponding subsections.
  - In *Section 10: Traffic Control Layouts – Mobile Work,* see the corresponding subsections.
  - Other references include layouts such as *Figure 6H-35 – Mobile Operation on a Multi-Lane Road* found in the U.S. Department of Transportation Federal Highway Administration's Manual on Uniform Traffic Control Devices
- In addition to meeting WorkSafeBC requirements for personal protective equipment, those working on Provincial roadways shall comply with *Section 5.4.3: Apparel for Other Onsite Workers.*

Guidance:

- For a recovery that does not pose a hazard to the travelling public but requires additional resources, it is important to secure and establish them before commencing the recovery operation. The requirements for this situation may include, but are not limited to:
  - additional traffic control signs and devices
  - Traffic Control Persons
  - special equipment
- A recovery that may encroach into the travel lane will probably require the closure of one or more lanes. This includes situations where:
  - The tow truck or disabled vehicle is fully or partially blocking a lane.
  - Tow truck equipment—cables, outriggers, cranes, and other equipment—is crossing, encroaching on, or operating above the travel lanes.

Options:

- For all the layouts in Sections 7 to 18 where a Crew Working Ahead C-004 sign or a Construction Ahead C-018-1A sign is required, an Accident Scene, Emergency Scene, Emergency Incident, or Tow Truck (C-058, C-070, C-071 series) sign with flags or a flashing light may be used instead.

19.2 Roadway Closure – No Through Road – Local Road

Purpose:

This layout shows a typical setup for a long duration local road closure where network street continuity is lost due to the closure. Advanced planning for detours is key to success as notifications, jurisdictional acceptance and/or the development of custom signs may be necessary. Emergency services (Police, Ambulance, Fire) may need to be contacted depending on the complexity of the closure (network of streets impacted), and overall length (time and distance) of detour.

Figure 19.2: Roadway Closure – No Through Road shows the general layout for a singular road closure.

Standard:

- All detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour signs shall be in place along the corresponding detour route.
- Barricades shall span the entire width of the roadway.

Guidance:

- Regulatory traffic control devices should be added or modified as needed for the duration of the detour.
- Additional signs should be erected at all connecting roadways to provide clear guidance to alternative routes for connectivity.

Options:

- If the road is opened for some distance beyond the intersection and/or there are significant origin/destination points beyond the intersection such as residences, the Road Closed Local Traffic Only C-201 series sign and barricades may be located at the edge of the traveled way to allow local passage.
- A Checkerboard C-114 sign may be placed at the closure, just in front of, or as part of the barricade sign assembly. If the C-114 sign is used, the Road Closed R-012 sign may be mounted on the barricade or below the C-114 sign. The C-114 sign provides for better notification and stop compliance.
- If the road closure is some distance from the intersection, the Road Closed Ahead C-030-6A sign may be added to provide additional warning of the closure ahead.
- DMS messages may be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

FIRST AVE CLOSED	BEGINS APRIL 20
ROAD CLOSED	DETOUR MAIN ST

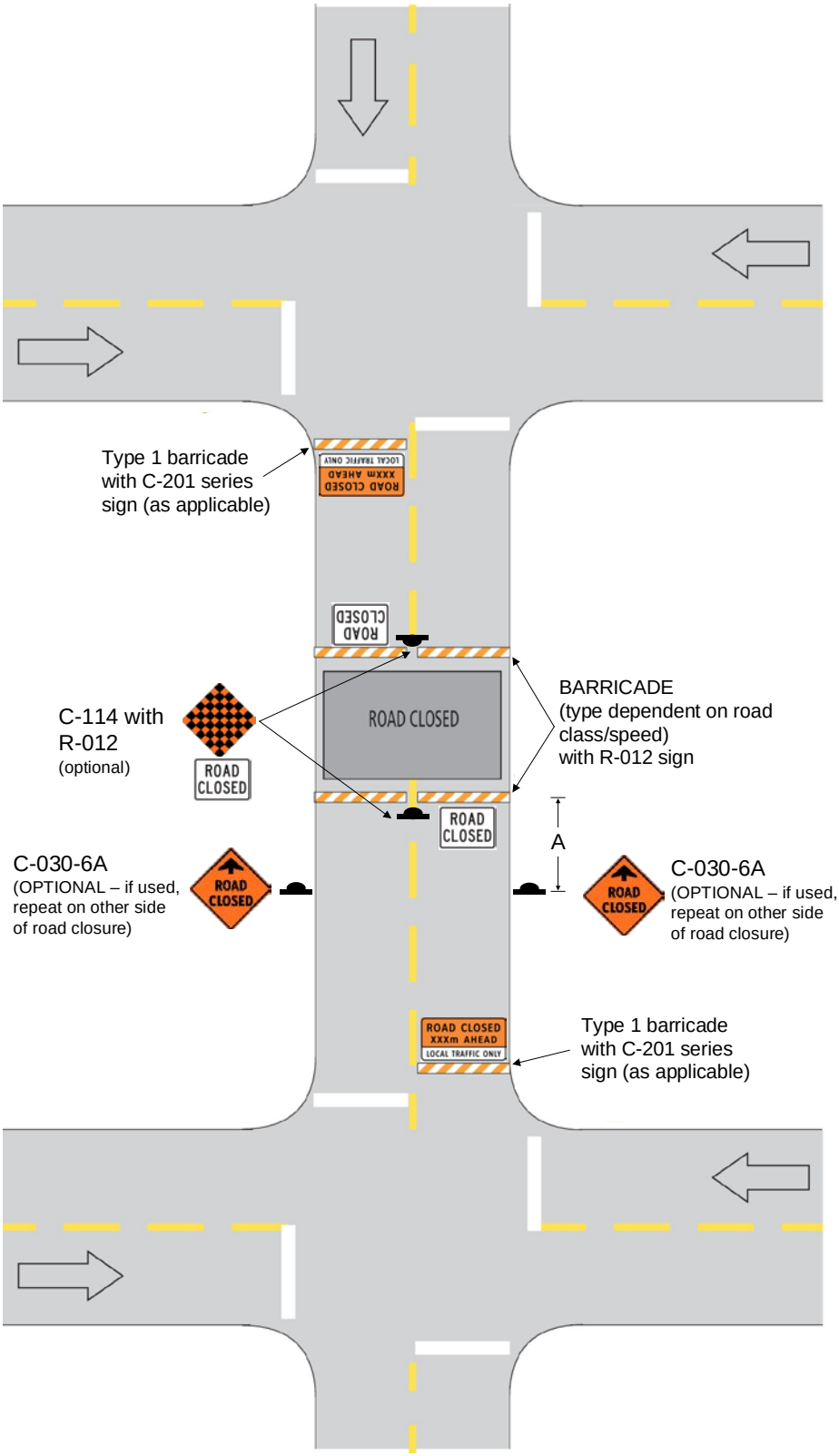
- If the road closure impacts a road commonly used to access a highway or a popular destination, additional signage may be required to establish a detour.
- The “LOCAL TRAFFIC ONLY” text of the Road Closed Local Traffic Only C-201 series signs may be replaced with custom messaging such as road closure times or dates to provide more information representing local conditions to the road user.
- Typical custom signage used in detours include the following examples:





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Figure 19.2: Road Closure – No Through Road – Local Road



19.3 Roadway Closure – Two-Lane, Two-Way Roadway

Purpose:

This layout shows a typical setup for a long duration road closure on two-lane, two-way roadways. Advance planning for road closures and detours is key to success as notifications, jurisdictional acceptance and/or the development of custom signs may be necessary. Emergency services (Police, Ambulance, Fire) may need to be contacted depending on the complexity of the closure (network of streets impacted), and overall length (time and distance) of detour.

Figure 19.3: Roadway Closure – Two-Lane, Two-Way Roadway shows the general layout for a two-lane, two-way road closure.

Standard:

- All road closures and detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour or U-turn signs shall be in place along the corresponding detour or U-turn route.
- Barricades (typically type 3) shall span the entire width of the roadway.

Guidance:

- Regulatory traffic control devices should be added or modified as needed for the duration of the closure.
- Speed limits for road closures should be established at 50 km/h.
- Advanced information should be provided. This information can be provided by several means:
    1) Portable DMS - messages at major intersections prior to the closure location
    2) Permanent Overhead DMS managed by the Transportation Management Centre of BC (TMCBC) – closure messages may be displayed up to several hundred kilometres from the closure location, depending on availability.
    3) Static Signs – custom messages at major intersections prior to the closure location
    4) DriveBC – all closures should be posted online
    5) Social media, radio, television (as required based on the complexity and duration of the situation)

Options:

- DMS messages may be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

HWY X CLOSED XX KM AHEAD	AUTHORIZED VEHICLES ONLY
HWY X CLOSED AT MAIN ST	WATCH FOR SIGNS USE U-TURN

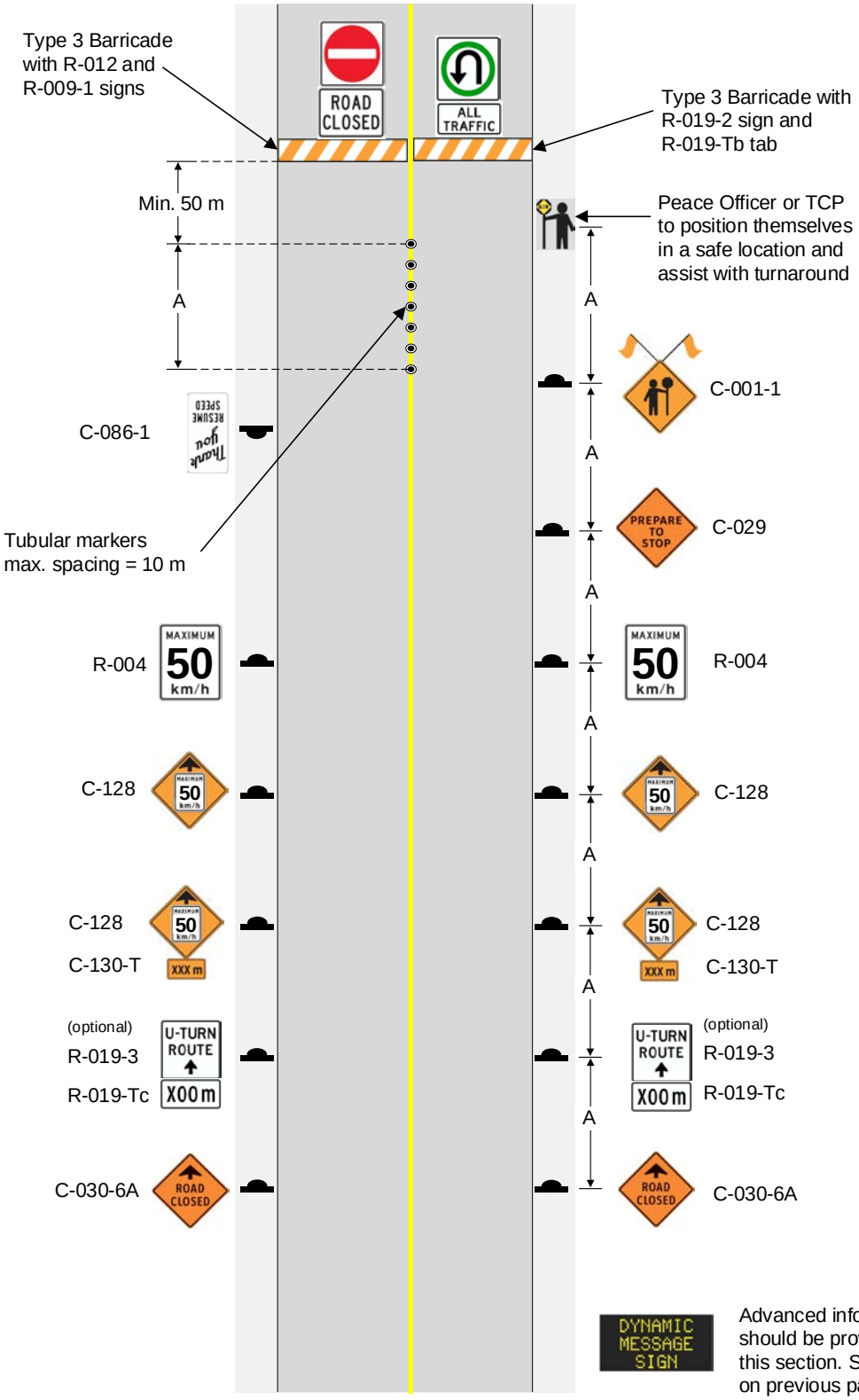
- A Checkerboard C-114 sign may be placed at the closure, just in front of, or as part of the barricade sign assembly. If the C-114 sign is used, the Road Closed R-012 sign may be mounted on the barricade or below the C-114 sign. The C-114 sign provides for better notification and stop compliance.
- The U-turn Route Ahead R-019-3 sign with R-019-Tc distance tab may be used to provide additional warning to drivers that there is an established U-turn location ahead.
- If the road closure impacts a road commonly used to access a highway or a popular destination, additional signage may be required to establish a detour.
- Typical custom signage includes the following examples:





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Figure 19.3: Roadway Closure – Two-Lane, Two-Way Roadway



19.4 Roadway Closure – Multilane Roadway

Purpose:

This layout shows a typical setup for a long duration road closure on a multilane roadway. Advance planning for road closures and detours is key to success as notifications, jurisdictional acceptance and/or the development of custom signs may be necessary. Emergency services (Police, Ambulance, Fire) may need to be contacted depending on the complexity of the closure (network of streets impacted), and overall length (time and distance) of detour.

Figure 19.4: Roadway Closure – Multilane Roadway shows the general layout for a multilane roadway.

Standard:

- All road closures and detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour or U-turn signs shall be in place along the corresponding detour or U-turn route.
- Barricades (typically type 3) shall span the entire width of the roadway.

Guidance:

- Regulatory traffic control devices should be added or modified as needed for the duration of the closure.
- Speed limits for road closures should be established at 50 km/h.
- Advanced information should be provided. This information can be provided by several means:
    1) Portable DMS - messages at major intersections prior to the closure location
    2) Permanent Overhead DMS managed by the Transportation Management Centre of BC (TMCBC) – closure messages may be displayed up to several hundred kilometres from the closure location, depending on availability.
    3) Static Signs – custom messages at major intersections prior to the closure location
    4) DriveBC – all closures should be posted online
    5) Social media, radio, television (as required based on the complexity and duration of the situation)

Options:

- DMS messages may be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

HWY X CLOSED XX KM AHEAD	AUTHORIZED VEHICLES ONLY
HWY X CLOSED AT MAIN ST	WATCH FOR SIGNS USE U-TURN

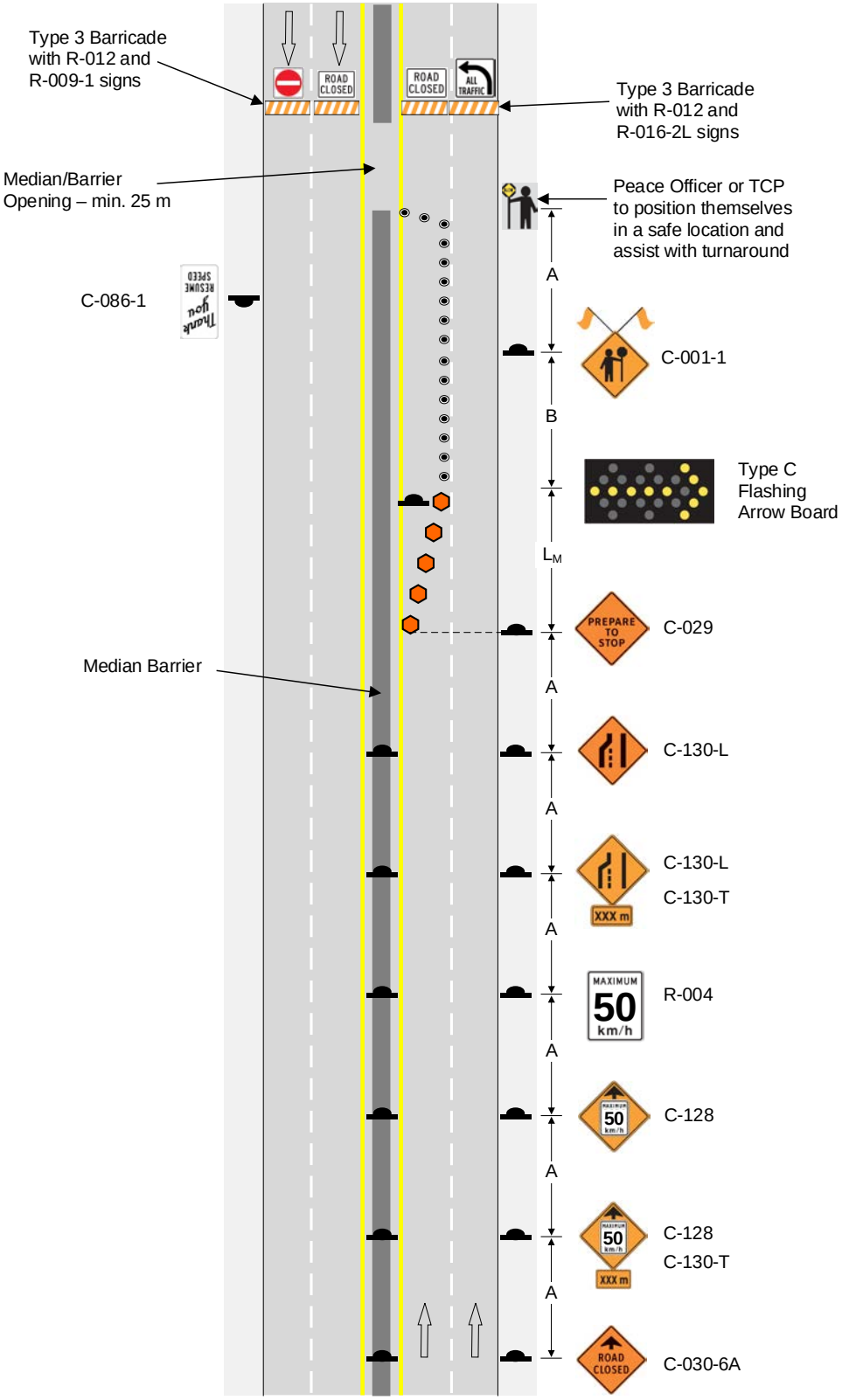
- A Checkerboard C-114 sign may be placed at the closure, just in front of, or as part of the barricade sign assembly. If the C-114 sign is used, the Road Closed R-012 sign may be mounted on the barricade or below the C-114 sign. The C-114 sign provides for better notification and stop compliance.
- The U-turn Route Ahead R-019-3 sign with R-019-Tc distance tab may be used to provide additional warning to drivers that there is an established U-turn location ahead.
- If the road closure impacts a road commonly used to access a highway or a popular destination, additional signage may be required to establish a detour.
- Typical custom signage includes the following examples:





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Figure 19.4: Roadway Closure – Multilane Roadway



19.5 Roadway Closure – Intersection (Signalized or Unsignalized)

Purpose:

This layout shows a typical setup for a long duration road closure at signalized or unsignalized intersections. Advance planning for road closures and detours is key to success as notifications, jurisdictional acceptance and/or the development of custom signs may be necessary. Emergency services (Police, Ambulance, Fire) may need to be contacted depending on the complexity of the closure (network of streets impacted), and overall length (time and distance) of detour.

Figure 19.5: Roadway Closure – Intersection (Signalized or Unsignalized) shows the general layout for a road closure at signalized or unsignalized intersections.

Standard:

- All road closures and detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour or U-turn signs shall be in place along the corresponding detour or U-turn route.
- Barricades (typically type 3) shall span the entire width of the roadway.

Guidance:

- At signalized intersections, traffic signals should be put into flash mode, turned off, or covered/bagged.
- Regulatory traffic control devices should be added or modified as needed for the duration of the closure.
- Speed limits for road closures should be established at 50 km/h.
- Advanced information should be provided. This information can be provided by several means:
    1) Portable DMS - messages at major intersections prior to the closure location
    2) Permanent Overhead DMS managed by the Transportation Management Centre of BC (TMCBC) – closure messages may be displayed up to several hundred kilometres from the closure location, depending on availability.
    3) Static Signs – custom messages at major intersections prior to the closure location
    4) DriveBC – all closures should be posted online
    5) Social media, radio, television (as required based on the complexity and duration of the situation)

Options:

- DMS messages may be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

HWY X CLOSED XX KM AHEAD	AUTHORIZED VEHICLES ONLY
HWY X CLOSED AT MAIN ST	WATCH FOR SIGNS USE U-TURN

- A Checkerboard C-114 sign may be placed at the closure, just in front of, or as part of the barricade sign assembly. If the C-114 sign is used, the Road Closed R-012 sign may be mounted on the barricade or below the C-114 sign. The C-114 sign provides for better notification and stop compliance.
- If the road closure impacts a road commonly used to access a highway or a popular destination, additional signage may be required to establish a detour.
- Typical custom signage includes the following examples:





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